

ACPE 2026

ISPE'S 18TH ASIAN CONFERENCE ON PHARMACOEPIDEMIOLOGY

30 OCT - 1 NOV 2026 MELBOURNE, AUSTRALIA

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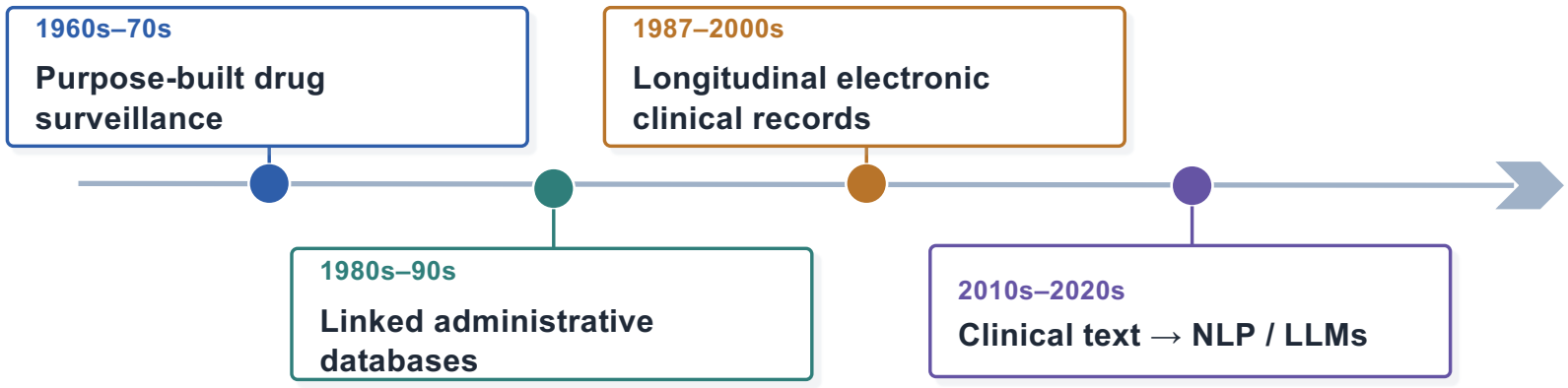
More data, more biases?

Treading carefully while designing NLP and LLM-enabled pharmacoepidemiology research

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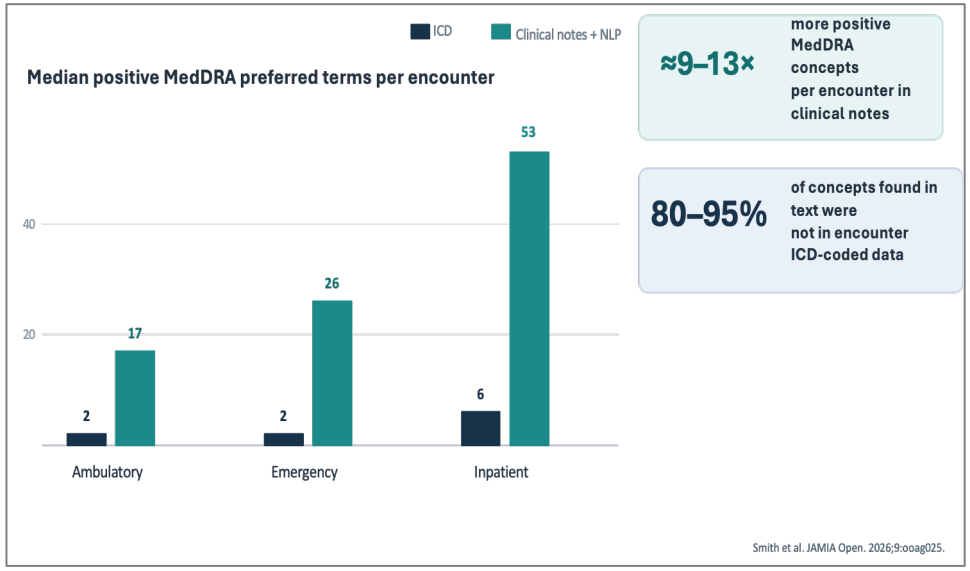


NLP/LLM: Next step in long history of novel data sources in pharmacoepidemiology

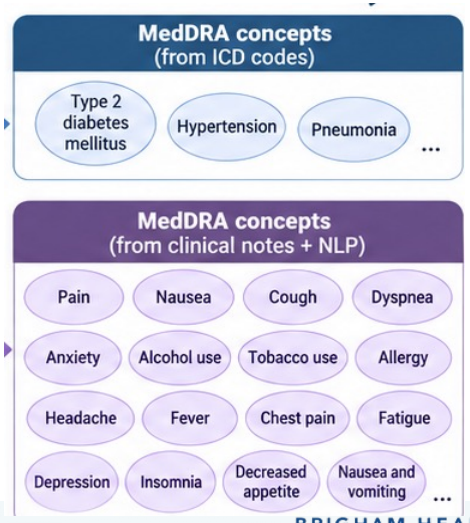


NLP/LLM extracts substantially more medical concepts than structured data

Vanderbilt University Medical Center, covering 2010–2023.



Clinical notes frequently contained symptoms that might not warrant a separate diagnosis code.



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NLP/LLM can do TTE!!

Table 1
A summary of the target trial framework and how natural language processing (NLP) techniques can help fulfill each component.

Protocol component	Description	NLP use	Example		
			Author	Application	Challenge
Eligibility criteria and follow-up period	Cohort and time-zero identification after applying target trial eligibility criteria	Extract inclusion and exclusion criteria from free text	[7]	To identify MM patients on AR treatment, a key inclusion criterion	Details of MM treatment often recorded in clinical notes
Treatment strategies	Well-defined treatments being compared	Extract data from free text required to designate treatment strategy	[8]	To identify those with and without a history of SCD, the exposure of interest	SCD is diagnosed via neuroimaging reports as clinical notes
			[9]	To identify patient suicidal ideation profile, the exposure of interest	Suicidal ideation does not have a place as a discrete variable
			[10]	To extract dose number, dose frequency, and dose interval from free text to calculate exposure duration	Prescription instructions often contain variability preventing the use of discrete fields
Assignment procedures	Random assignment to each treatment strategy	Identify important confounding variables from the free text	[11]	To identify characteristics to define cancer staging, a confounding variable	Details to determine cancer staging are found in oncology notes
			[12]	To identify numerous SDoH, confounding variables	SDoH often do not have a place as discrete variables
Outcome of interest	Outcome being measured and compared between groups	Identify the occurrence of the outcome if recorded in the free text	[13]	To identify arthralgia occurrence, the outcome of interest	There are inconsistencies in how arthralgia is recorded by gastroenterologists in the EHR
			[14]	To identify cases of shoulder injury related to vaccine administration (SIRVA)	There are no defined diagnosis codes for SIRVA to record in structured text
Causal contrasts of interest and analytic effect	Desired causal effect to be estimated (i.e intention to treat effect and per-protocol effect) and plan to validly estimate the effect	Extract information on adherence to treatment strategy and post-baseline confounding variables from the free text	Hypothetical	To identify HIV treatment compliance to estimate a per-protocol effect	HIV patients miss appointments making it challenging to determine compliance which may be recorded in the free text
			Hypothetical	To identify cancer staging, a time-varying confounding variable	Details to determine cancer staging are found in oncology notes

https://doi.org/10.1016/j.gloepi.2025.100204

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Categorizing the biases!

- Biases exclusive to NLP methods
- Biases similar to structured data studies
- Meta-research concerns

Challenges associated with NLP/LLM-based measurement

Annotation complexities

Learning to detect and understand drug discontinuation events from clinical narratives

Feifan Liu,[1] Richeek Pradhan,[1] Emily Druhl,[2] Elaine Freund,[1] Weisong Liu,[3] Brian C. Sauer,[4] Fran Cunningham,[5] Adam J. Gordon,[6,7] Celena B. Peters,[4,6] and Hong Yu[2,3,8,9]

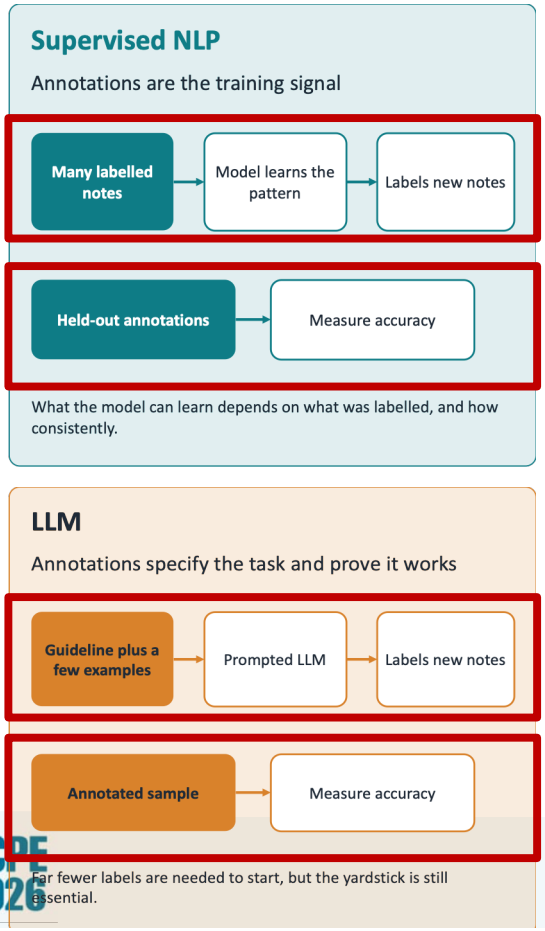
DDC reason category	Scope of coverage
Adverse_Event	Adverse drug events leading to a DC, such as side effects, contraindication, toxicity, drug interactions, etc.
Drug_Modification	Non-AE related drug modification demanded by physician, such as drug switch due to ineffectiveness, or procedure accommodation, etc.
Non_Medical	This category covers natural stop cases (eg, drug prescribed is finished or not needed) or patient self-discontinuation (eg, nonadherence or formulary change)
Unknown	No reason is mentioned

Her Bactrim was discontinued and her rash is slowly resolving

Diabetes was uncontrolled and patient is taken off metformin due to worsening renal function and started on Novolog and Lantus.

I have asked him to discontinue his Percocet as he is exceeding the Tylenol limitations per day.

Annotation complexities



Biased or inconsistent model training

Uncertain sensitivity and specificity

Enriched annotation datasets

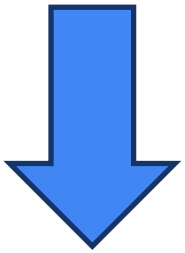
Liu et al., 2019

We collected 1,867 clinical notes from patients.

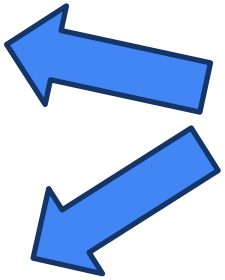
100,151 sentences.
- 1,978 sentences with drug discontinuation events.
- Approximately 2% positive sentences.

??

- Inefficient annotation
- Insufficient positive examples



Key word-based enrichment



Oversampling for training

- Subtle, unusual or differently phrased cases are underrepresented
- Subgroup specific performance

Oversampling in validation data

- Validation metrics dependent on prevalence might be highly affected

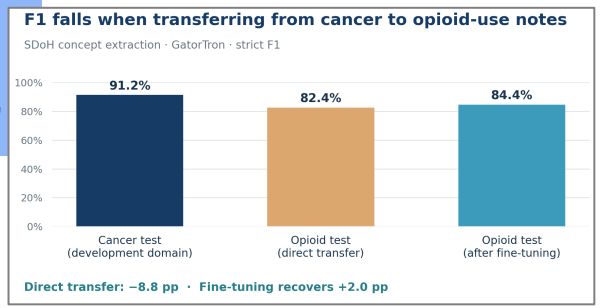
Enriched annotation datasets: examples

Identifying Social Determinants of Health from Clinical Narratives: A Study of Performance, Documentation Ratio, and Potential Bias

Yu et al. J Biomed Inform. 2024;153:104642.

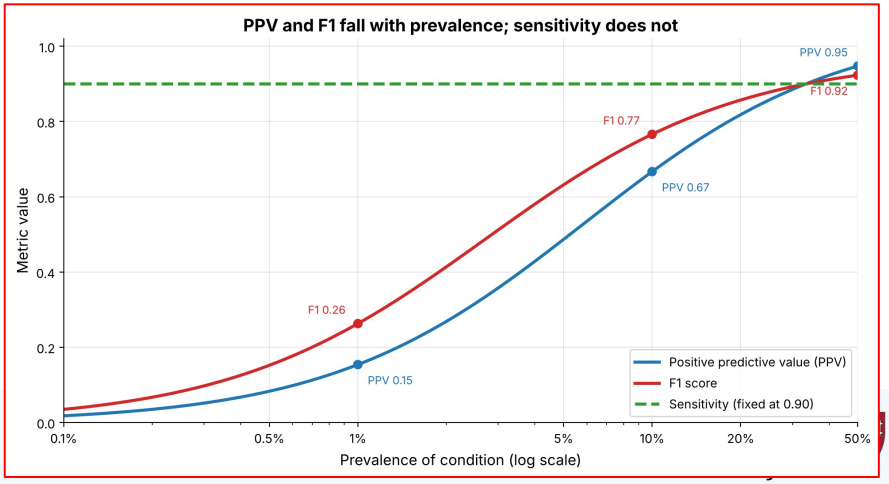
Pretrained, encoder-based transformer language models

"Then, we identified clinical notes with at least three unique mentions of SDoH keywords and randomly sampled a subset for annotation. After annotation, we divided the annotated notes into a training set and a testing set with an 8:2 ratio and **held out 10% of the training sample which we used as a validation set.**"



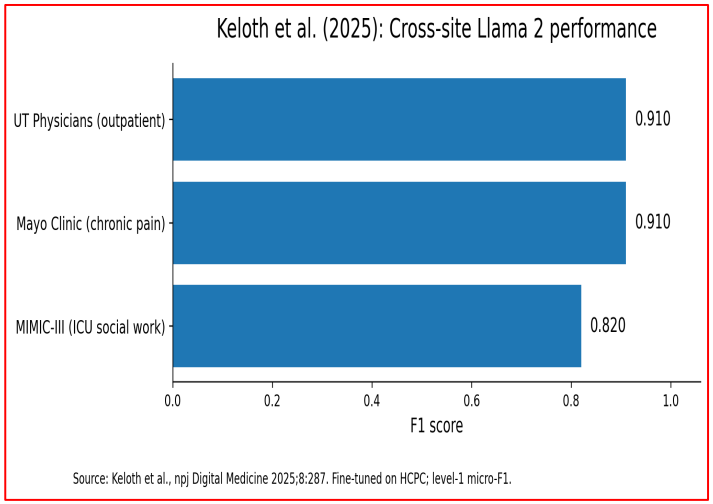
Nature of metrics

Prevalence	Diseased	TP	FN	FP	TN	PPV	F1
50%	5,000	4,500	500	250	4,750	0.947	0.923
10%	1,000	900	100	450	8,550	0.667	0.766
1%	100	90	10	495	9,405	0.154	0.263

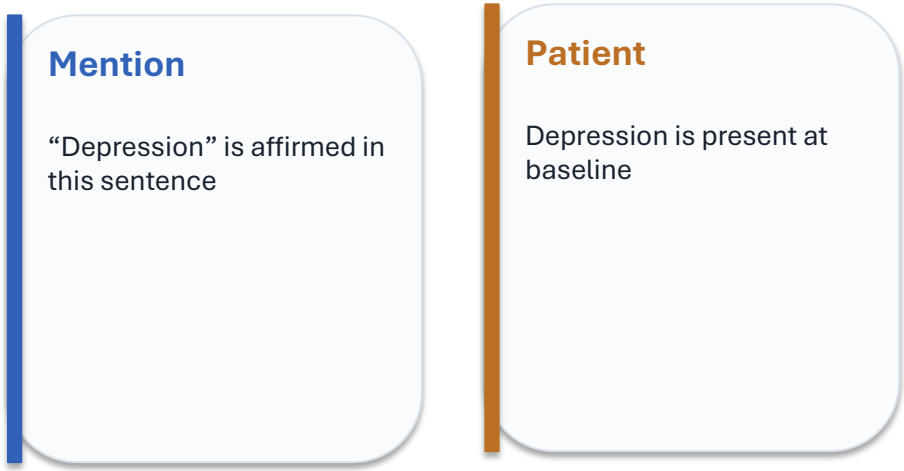


Transportability across hospitals, populations and note types

	Keloth et al. (2025)
Task	Sentence-level extraction of 21 SDOH factors
Model	Fine-tuned Llama 2
Development setting	Harris County Psychiatric Center, Texas
Other settings	UT Physicians, Mayo Clinic, MIMIC-III
What differed?	Patient populations, clinical specialties, note content and SDOH distributions
Main finding	The same LLM achieved different F1 scores across receiving datasets



Concept extraction versus patient-level phenotype validity

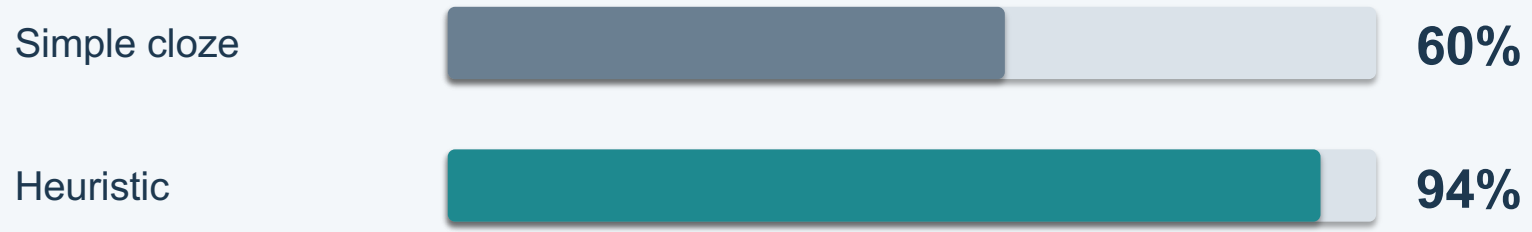


91.9% mention-level accuracy ≠ 91.9% patient-level phenotype accuracy.

Prompt dependence, model versioning and implications for quantitative bias analysis

Same model (GPT-3.5) • Same task (clinical coreference resolution)

Classification accuracy by prompting strategy



Sensitivity and specificity may vary across settings

Misclassification parameters may be specific to the measurement context

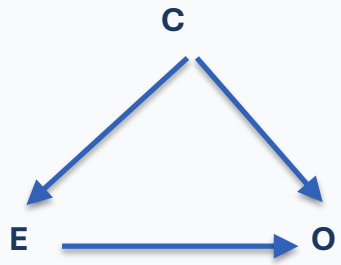
$$Se_{a,h,p,m} \quad , \quad Sp_{a,h,p,m}$$

- **a** Treatment / exposure group
- **h** Hospital or healthcare setting
- **p** Prompt used for classification
- **m** LLM model or version

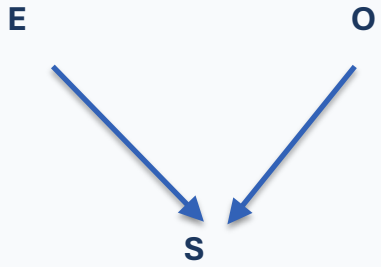
Quantitative Bias Analysis may require setting-specific sensitivity and specificity—not one common pair.

Category 2: Familiar epidemiologic biases that *may* operate differently with NLP

CONFOUNDING



SELECTION



Condition on S = 1

INFORMATION



Why are clinical notes written in the first place?

1
Clinical state
What is actually true of the patient

2
Healthcare contact
Does the patient seek or receive care?

3
Assessment
Does the clinician ask about the construct?

4
Documentation
What is written, and what is copied forward?

5
Capture and linkage
Is the note inside the database you can see?

6
Note selection
Which notes does the researcher send to the model?

7
NLP or LLM extraction
Concept, negation, timing, certainty, experiencer

8
Patient-level phenotype
The researcher's rule that aggregates mentions

Latent assumptions in ICD codes versus the additional contextual capabilities of NLP/LLMs

Context	Clinical note + NLP	ICD-coded diagnosis
Assertion	Present / negated / hypothetical	Present
Temporality	Current / historical / future	Encounter date; some history codes
Experiencer	Patient / family member / other	Usually the patient
Certainty	Confirmed / probable / possible	Moderate to high

Informative presence: opportunity to be documented

Even with identical true disease states, classification may depend on encounters.

Patient A — 2 visits

Few opportunities for documentation or erroneous NLP detection.

Patient B — 25 visits

More opportunities for both true detection and false positives.

Illustration: if note-level specificity = 99% and errors are independent,

patient specificity after 25 notes = $0.99^{25} \approx 77.8\%$

Both NLP and ICD can be affected; which suffers more is an empirical question.

McGee et al., Epidemiology 2022. https://pmc.ncbi.nlm.nih.gov/articles/PMC8633193/

More sensitive is not necessarily more specific

Comparisons with chart review show phenotype-dependent trade-offs.

Symptom	ICD specificity	NLP specificity
Fatigue	98%	77%
Depression	99%	81%
Pain	95%	11%

These are study- and phenotype-specific results—not a universal NLP disadvantage.

Hemodialysis symptom study (2020): https://pmc.ncbi.nlm.nih.gov/articles/PMC7001114/

Multiple mentions ≠ multiple independent assessments

Repeated documentation can create false confidence in an NLP phenotype.

Two separate assessments

Visit 1: “Depression currently active.”

Visit 2: “Persistent depressive symptoms.”

Copied-forward documentation

Visit 1: “Depression currently active.”

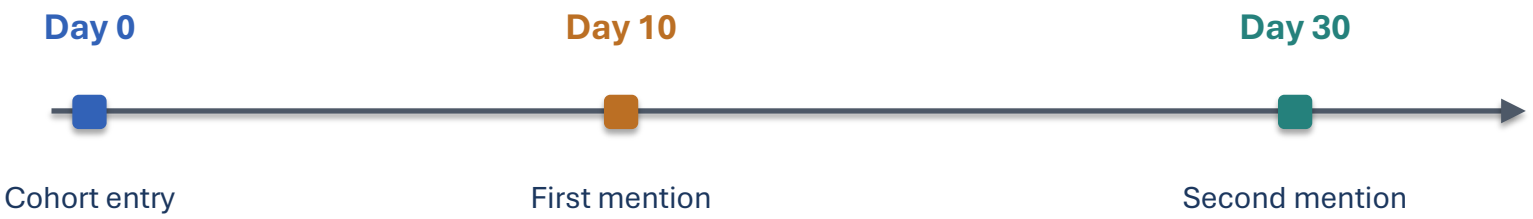
Visit 2: “Depression currently active.” (copied)

Repeated text is not equivalent to independent clinical confirmation.

Steinkamp et al. Prevalence and Sources of Duplicate Information in the EMR. JAMA Netw Open 2022.

When the phenotype requires looking into the future

Two positive mentions within 30 days may be used to classify an earlier event.



Three different concerns

Delayed ascertainment · Temporal misclassification · Future-dependent selection/classification

Selection: whose notes are available and processed?

Choosing notes and choosing patients create different problems.

Design decision	What may go wrong?
All patients included; psychiatry notes processed	Exposure/outcome may be measured differently across patients
Only patients with psychiatry notes included	Study population is selected on care/documentation
Care outside the EHR not linked	Relevant clinical information is unobserved

Not every note-restriction problem is selection bias; many are measurement problems.

Weiskopf & Weng, JAMIA 2013; Goldstein, Curr Epidemiol Rep 2025.

NLP can recover missing confounders

But information availability may itself depend on the treatment or patient.

- Depression severity, substance use, functional status and social context may be found in notes.
- NLP can supplement structured data and potentially reduce residual confounding.
- Yet documentation is driven by specialty, utilization and clinical workflow.

Does NLP improve exchangeability equally across treatment groups?

Rafalko et al., Global Epidemiology 2025;9:100204.

Differentially missing confounders

Accurate extraction from existing notes does not remove unrecorded information.

Baseline depression severity	Treatment A	Treatment B
Documented	80%	35%
Not documented	20%	65%

Hypothetical example

- Is recording associated with actual severity?
- Does the absence of a mention mean the absence of severity?
- What are the implications for propensity-score adjustment?

Weberpals et al., 2024: https://pmc.ncbi.nlm.nih.gov/articles/PMC11127690/; Blake et al., 2020.

Category 3: Meta-research considerations

AI USE IN PUBLIC HEALTH AJPH

Associations Between Natural Language Processing–Enriched Social Determinants of Health and Fatal Opioid Overdose Among US Veterans, Fiscal Year 2018–2021

Avijit Mitra, PhD, Richeek Pradhan, MD, PhD, Kirsha S. Gordon, PhD, Haijuan Yan, PhD, Robert D. Kerns, PhD, William C. Becker, MD, and Hong Yu, PhD

See also AI in Public Health, p. 1484–1574.

Economizing: selecting high-yield notes, computational sustainability, privacy and governance

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SDOH factors	NLP-Extracted, AOR (95% CI)[a]	Structured, AOR (95% CI)[a]	Inclusive, AOR (95% CI)[a]	Conservative, AOR (95% CI)[a]
Social problems[b]	1.86 (1.69, 2.05)	1.76 (1.48, 2.09)	1.85 (1.67, 2.04)	1.80 (1.51, 2.14)
Financial problems[c]	2.18 (1.99, 2.40)	2.42 (2.14, 2.74)	2.20 (2.00, 2.43)	2.63 (2.32, 2.98)
Housing instability	2.35 (2.14, 2.59)	2.99 (2.64, 3.38)	2.32 (2.10, 2.55)	3.36 (2.96, 3.81)
Legal problems	2.60 (2.35, 2.87)	2.81 (2.35, 3.37)	2.59 (2.34, 2.86)	3.07 (2.55, 3.70)
Violence	2.05 (1.86, 2.25)	2.13 (1.90, 2.38)	2.08 (1.89, 2.29)	2.47 (2.18, 2.80)
Barriers to care	2.19 (1.99, 2.40)	. . .	2.15 (1.95, 2.36)	. . .
Transition of care	1.93 (1.72, 2.17)	. . .	1.90 (1.69, 2.14)	. . .
Food insecurity	1.78 (1.59, 1.99)	. . .	1.74 (1.56, 1.96)	. . .
Nonspecific psychosocial needs	. . .	1.72 (1.52, 1.96)	1.69 (1.49, 1.91)	. . .

Economizing: selecting high-yield notes, computational sustainability, privacy and governance

Appendix E: SDOH Occurrences According to Data Source

SDOH	Unstructured data - NLP (%)	Structured data (%)	Present in both (%)
As Covariates			
Social problems	99.6	9.2	8.7
Financial insecurity	97.6	29.7	27.3
Housing insecurity	97.2	36.0	33.2
Legal problems	98.5	18.8	17.3
Violence	93.5	30.9	24.4
Macro Average	97.3	24.9	16.2
As Exposures			
Social problems	99.5	9.2	8.7
Financial insecurity	97.1	30.5	27.7
Housing insecurity	96.7	36.9	33.6
Legal problems	98.6	18.1	16.7
Violence	92.3	31.8	24.2
Macro Average	96.9	25.3	22.2

Thinking through whether NLP/LLM improve the epidemiologic answer?

Evaluate incremental validity—not simply the number of extracted concepts.

Research question	Evidence to examine
Additional true cases?	Patient-level sensitivity and specificity
Different effect estimate?	ICD-only, NLP-only and combined analyses
Less residual confounding?	Confounder completeness and error
Transportable findings?	Subgroup and external validation
Worth the added complexity?	Compute, privacy, annotation, reproducibility